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The Long-Term Direct and External Effects of Jewish Expulsions in Nazi Germany†

By Mevlude Akbulut-Yuksel and Mutlu Yuksel*

This paper examines the long-term direct and spillover effects of large-scale human capital loss caused by the persecution of Jewish professionals in Nazi Germany. Using region-by-cohort variation in the percentage of the Jewish population as a quasi-experiment, we find that German children who were at school-age during the persecutions have fewer years of schooling on average in adulthood. Moreover, these children are less likely to finish high school and go to college. These results are robust after controlling for regional unemployment and income per capita, wartime destruction, Nazi and Communist Party support, compulsory schooling reform, migration, urbanization, and mortality. (*JEL* I21, I28, J24, J44, N34, N44, Z12)

An extensive literature on endogenous economic growth has emphasized the role of human capital as a source of long-run differences in income levels across countries. The loss of productive skills due to armed conflicts, ethnic cleansing, natural disasters, diseases, and brain drain immigration has become common during the last century (Blatmann and Miguel 2010; de Walque and Verwimp 2010; Carrington and Detragiache 1998). Evidence from macro-level studies suggests that the loss of high-skilled professionals has numerous adverse effects on a country's macroeconomic performance and growth prospects (Klenow and Rodríguez-Clare 2005; Lucas 1988). More importantly, the lack of highly skilled professionals leaves children and young adults vulnerable, given the age-specific aspect of many human capital investments.

This paper provides causal evidence on long-term direct and spillover effects of the loss of high-skilled professionals on children's outcomes. Specifically, we analyze the long-term consequences of the persecution of Jewish professionals in Nazi

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Germany on German children's human capital formation. Hitler's Nazi Party passed the "Law for the Restoration of the Professional Civil Service" in April 1933 shortly after they came into power.[1] This law allowed the Nazi government to purge Jews from the civil service, a vast organization in Germany that included teachers, university professors, judges, and many other professionals. In May 1933, this law was amended to include the postal service, railroads, communal health insurance systems, professional associations, unemployment and salaried employees' insurance, miners' guilds, trade guilds, and chambers of commerce (Kaplan 2005). With the civil service law, more than 15 percent of university professors, teachers, doctors, lawyers, and other professionals who were Jewish were dismissed from their professions (Strauss 1983; German Municipalities Statistical Yearbooks 1933, 1935).[2] The dismissal of school principals, teachers, and professors might have adversely affected the educational decisions of German children. Similarly, declining teacher quality and bigger class sizes could have increased the number of early dropouts. In addition, the systematic persecution of the Jewish population might have changed the German children's peers in school, led to less competition in the labor market and caused psychological effects on German children due to visible unrest, violence in the streets and loss of peers and friends.

There was a significant regional variation in the extent of dismissals within Germany. On the one hand, areas with a higher fraction of Jewish population were exposed to a dramatic change in the number of teachers, professors, and other professionals compared to regions with a lower fraction of Jews.[3] On the other hand, only individuals who were of school-age during the Nazi Regime would have had their human capital formation affected by the exile of Jewish professionals. This paper therefore uses a difference-in-differences-type strategy where the "treatment" variable is an interaction between the percentage of the population that is Jewish in 1933 and a dummy variable for being school-aged during the Nazi Regime where we always control for region and birth year fixed effects.

We find that the expulsion and exile of Jewish professionals had enduring detrimental effects on German children's educational attainment. Children who were school-aged during the persecutions attained 0.5 fewer years of schooling on average. These children were also less likely to finish high school and attend a technical school or university. More significantly, we find that the dismissal of Jewish professors and schoolteachers had a sizable adverse effect on German children's education decision.

During the time period we focus on in our paper, Germany also experienced additional political and economic events. The nationwide effects of these events are captured by the birth year fixed effects in our analysis. We also use a lower level of geographical aggregation than state in our estimation, which allows us to account for potential state-specific events and policies. In addition, we formally test whether

[1] In German, "Gesetz zur Wiederherstellung des Berufsbeamtentums."

[2] Starting from 1933, many Jewish professors including Albert Einstein, Gustav Hertz, Erwin Schroedinger, Max Born as well as twenty past and future Nobel Prize winners left the country (Evans 2005). Similarly, German Municipalities Statistical Yearbooks report that 8.3 percent of teachers were dismissed by the Nazi government in 1933.

[3] Figure 3 and Figure 4 show that regions with a higher percentage of ex ante Jewish population had a greater decline in the numbers of science professors and teachers after the Nazi Party came into power in 1933.

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the affected cohorts were differentially affected by these events in Section IV. We find that long-term adverse effects of the Jewish expulsions remain economically and statistically significant even after we account for regional unemployment and income per capita, wartime destruction, the timing of the compulsory schooling reform, migration, mortality and urbanization. Also, our results are robust to the different categorization of the affected and the control cohorts.

This paper makes several contributions. First, our study contributes to the broader literature exploring the association between large negative shocks and children's human capital formation. This strand of literature finds that armed conflicts, natural disasters, political instability, and macroeconomic crises have long-lasting, detrimental effects on children's educational attainment (Akbulut-Yuksel 2014; Meng and Gregory 2007; Ichino and Winter-Ebmer 2004). Blattman and Miguel (2010) point out that the rigorous research on long-term effects of forced high skilled emigration caused by such events is very limited. Therefore, it is critical to understand the long-run effects of the loss of high-skilled professionals and the mechanisms through which it affects children and young adults in the turbulent and conflict-torn countries around the globe. Our paper adds to this literature by quantifying the long-term consequences of systematic persecution of Jewish professionals in Nazi Germany on children's human capital formation.

Our study also contributes to growing literature on the long-term economic and political legacies of the Holocaust (Grosfeld, Rodnyansky, and Zhuravskaya 2013; Pascali forthcoming; Acemoglu, Hassan, and Robinson 2011). Acemoglu, Hassan, and Robinson (2011) show that German occupation led to lower postwar growth, long-term GDP per capita, and lower average wages in Russian cities and regions with a higher prewar Jewish population. Grosfeld, Rodnyansky, and Zhuravskaya (2013) and Pascali (forthcoming) provide further evidence on the long-term legacies of the Holocaust in Europe. Using variation in the "Pale of Settlement," Grosfeld, Rodnyansky, and Zhuravskaya (2013) find that cities in the Pale of Settlement have stronger support for socialist antimarket parties, lower rates of entrepreneurship and higher levels of trust today. Similarly, Pascali (forthcoming) documents that Italian cities with large Jewish communities in the fifteenth century currently have complex bank institutions.

Our paper adds to these recent studies by estimating the individual-level effects of the Holocaust. In addition, we provide formal evidence on the potential channels through which the purge of Jews affected human capital formation in Germany, such as a loss of professors and teachers. Our final contribution is to combine the newly assembled historical data on regional characteristics and the fraction of Jewish population in Nazi Germany with individual-level data from the German Socio-Economic Panel (SOEP), which allows us to control for individual and family background characteristics in our analysis in addition to regional characteristics.

This paper is closely related to two recent studies by Waldinger (2010, 2012). Waldinger (2012) finds that the productivity of peer scientists in the departments of dismissed Jewish professors remained virtually unchanged after the dismissals. On the other hand, Waldinger (2010) shows that in mathematics departments, the dismissal of Jewish professors had negative long-term consequences for the PhD students who graduated from these departments. The persecution of Jews from all professions and the associated loss of human capital in Germany at large, however,

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are likely to have more pervasive long-term effects beyond German Ph.D. students in mathematics. In our study, we take the potential spillover effects arising from the loss of high skilled professionals into account; hence, we estimate the long-term legacies of systematic expulsion of the Jewish population on the entire German population.

The remainder of the paper is organized as follows. Section I provides a brief background of the purge of Jewish professionals during the Nazi Regime. Section II discusses the identification strategy. Section III describes the data used in the analysis. Section IV presents the main results, extensions, and robustness checks. Section V concludes.

I. Background on Dismissal of Jewish Professionals during the Nazi Regime

The 1933 census documented that there were 525,000 Jews in Deutsche Reich at that time, composing 0.8 percent of the country's population (Friedländer 1998, Evans 2005). Although Jews comprised approximately one percent of the population in Deutsche Reich, they had been remarkably successful in German society and culture since their emancipation from legal restrictions in the course of the nineteenth century (Evans 2005). The Jewish community gained visibility by gradually concentrating in academia and in professions such as medicine, teaching, law, journalism, banking and finance in large numbers (Kaplan 2005).

The systematic persecution of the Jews began shortly after Hitler's Nazi Party came into power in 1933. Two turning points took place in the first few months of the Nazi Regime: the boycott of Jewish stores on April 1, 1933, and the exclusion of Jews from the civil service through the "Law for the Restoration of the Professional Civil Service" on April 7 (Evans 2005). This law allowed the Nazi government to purge Jews from civil service, a vast organization in Germany that included teachers, professors, judges, and many other professionals that were not government-controlled in other countries. In May 1933, the civil service law was amended to include supplemental regulations for other status groups, and the extensions of the term "civil service" to include a wide range of government positions as well as guilds (Kaplan 2005).

According to the "Law for the Restoration of the Professional Civil Service," a person was considered "non-Aryan" if he is descended from "non-Aryan," particularly Jewish parents or grandparents. It was sufficient if one parent or grandparent was of the Jewish faith (Yahil 1991). With this law, the Nazi government required all employees and officials in the public sector to present the "Aryan certificate" in order to identify the Jewish population (Heinemann 1999). The "Aryan certificate" encompassed the seven birth or baptism certificates (the person, his parents, and grandparents) and three marriage certificates (parents and grandparents) or certified proofs thereof. In the event that "Aryan" descent is questionable, the expert on racial research from the Ministry of Interior was asked to determine the individual's race/ethnicity (Yahil 1991).

As a result of the "Law for the Restoration of the Professional Civil Service," a growing number of Jewish teachers were forced to leave their jobs in the general school system. The German Municipalities Statistical Yearbooks report that 8.3 percent of teachers were dismissed by the Nazi government in 1933 in Germany;

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however, there was a substantial variation across German regions in the percentage of teachers that were dismissed. For example, 15 percent of male secondary school heads in Prussia and 32 percent of female secondary school heads in Berlin were fired in 1933. Similarly, 7.2 percent of male senior teachers were dismissed from their jobs in Prussia in 1933 (Evans 2005). The Jewish teachers who were not fired in April 1933, were forced into retirement without cause in 1935. Two years later, Jews and "half-Jews" were formally banned from teaching in non-Jewish schools. In 1936, there were 1,335 unfilled posts in elementary schools. The school system needed an additional 8,000 teachers a year to fill the open positions (Evans 2005). Due to the teacher shortage, the class size on average in all schools had increased to 43 students during this period, compared to 37 in 1927 (Evans 2005).

With the civil service law, Jewish professors in all fields were also dismissed. Approximately 15 percent (1,200 to 1,500) of university professors of all categories were dismissed and forced to emigrate from Germany after 1933 (Strauss 1983). When nonuniversity research scientists and scholars who had begun but not completed their training at the time of emigration are included, this figure becomes approximately 2,000. The replacement of the dismissed Jewish professors took years due to a shortage of qualified researchers as well as slow appointment procedures (Waldinger 2010, Evans 2005). Waldinger (2010) suggests that the positions of the dismissed professors could only be filled if the dismissed professors gave up their pension rights, because the dismissed were originally placed into early retirement. The states were reluctant to pay the salaries for the replacement and the pensions for the dismissed professors at the same time. Therefore, the reappointments took a long time.

Further legislation in 1933 sharply curtailed "Jewish activity" in the medical and legal professions. Within the same year, 16 percent of the lawyers lost their jobs because of the anti-Semitic persecutions (Jarausch 1986). In July 1933, Jewish doctors lost their patients with substitute health insurance schemes, and private insurance companies reimbursed the fees of Jewish doctors only for their Jewish patients. In 1938, all Jewish lawyers lost their admission to the bar and Jewish doctors lost their medical licenses. Removal of Jewish doctors from the profession created a large number of vacancies for German graduates to fill (Yahil 1991).

Taken together, Germany lost a substantial fraction of high-skilled professionals in all occupations within the civil service law. The severity of the human capital lost depended upon the fraction of the Jewish population residing in German regions before the expulsions began in 1933. The loss of Jewish educators and high-skilled professionals in a given region was plausibly exogenous conditional on controlling for fixed regional characteristics. Therefore, in our analysis, we will explore the cross-regional variation in the percentage of the population that is Jewish in 1933 to estimate the direct and spillover effects of Jewish expulsions on German children's human capital formation after we control for region fixed effects.

II. Identification Strategy

In this section, we describe our strategy for identifying the causal effect of the persecutions on the long-term educational outcomes of German children. This strategy exploits region-by-cohort variation in the fraction of Jewish population in Nazi

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Germany. This is a generalized difference-in-differences strategy where the "treatment" variable is an interaction between the percentage of Jews in German regions in 1933 and a dummy variable for being school-aged during the Nazi Regime.[4] In particular, the proposed estimate of the average treatment effect is given by β in the following baseline region and birth year fixed effects equation:

$$(1)\quad Y_{irt} = \alpha + \beta(JewishFraction_r \times AffectedCohort_{it}) + \delta_r + \gamma_t + \pi'\mathbf{X}_{irt} + \epsilon_{irt},$$

where Y_{irt} is the outcome of interest for individual i in region r born in year t. $JewishFraction_r$ is the percentage of the population that is Jewish in 1933 in region r. $AffectedCohort_{it}$ is a dummy variable that takes a value of 1 if individual i was born between 1910 and 1927 and zero otherwise. δ_r are region-specific fixed effects, controlling for the fact that regions may be systematically different from each other. γ_t are the birth year-specific fixed effects, controlling for the nationwide common shocks. $\mathbf{X}_{irt}$ is a vector of individual and household characteristics including gender and rural dummies and parental education. ϵ_{irt} is a random, idiosyncratic error term. The standard errors are clustered by region.

Individuals who were born between 1910 and 1927 form the affected cohorts since they were of primary school and college age in 1933 when the Nazi government started to force the Jewish out of the civil service, independent professions, and higher education. Thus, their schooling has the potential to be affected by the dismissal of Jewish professionals.[5] On the other hand, individuals born between 1951 and 1960 constitute the control group. These later birth cohorts attained their education after the postwar reconstruction was over in the late 1950s (Akbulut-Yuksel 2014); therefore, their educational attainment has not been affected by the persecution of Jewish professionals.[6] Individuals born between 1928 and 1950 were excluded from the analysis since their educational achievement has been affected by WWII and postwar reconstruction (Ichino and Winter-Ebmer 2004).

In order to interpret β as the effect of expulsions, we must assume that had persecutions not occurred, the affected cohorts and the cohorts born after WWII exhibit parallel trends in their educational achievement across regions of varying intensity of ex ante Jewish population. First, we assess the validity of the parallel trend assumption in Figure 1. More specifically, in Figure 1, we present the educational attainment by cohort in regions with a high and low fraction of Jewish population in 1933. For the analysis presented in Figure 1, we group the birth cohorts into five-year groups starting from 1900 in order to increase statistical precision. The vertical lines indicate our affected cohorts (i.e., born between 1910 and 1927) and the control cohorts (i.e., born between 1951 and 1960). This figure also helps us to

[4] This paper provides evidence on the long-term impacts of Jewish expulsions using region-by-cohort variation in the fraction of ex ante Jewish population within Germany; therefore, this approach may yield lower bound estimates for the aggregate nationwide effects of the persecution of Jews.

[5] The estimation results are robust to different age cutoffs and birth years of the affected cohorts, as reported in Appendix Table 3.

[6] An alternative control group could be individuals born before 1910. Estimation results where cohorts born before 1910 form the control group are presented in Appendix Table 4. We find quantitatively similar results to the baseline specification. Individuals born before 1910, however, are at 76 years of age and older in 1985, leading to small sample sizes.

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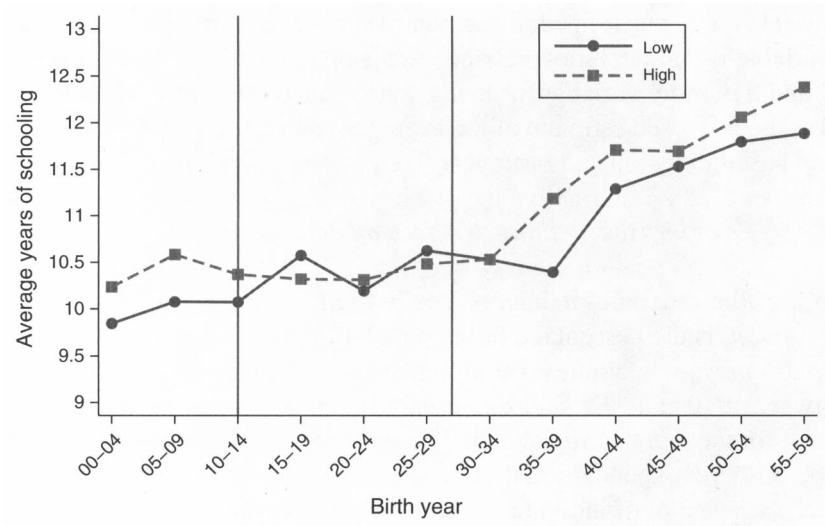

FIGURE 1. AVERAGE YEARS OF SCHOOLING

Source: Data is from the 1985 SOEP and authors' own calculations.

test whether our results are confounded by the differential postwar education policies. Germany, like many other countries, experienced an increase in educational attainment after WWII. Hence, the educational achievement was higher in urban areas before WWII; the rural areas had more scope to increase their educational achievement in the postwar era. Jews were more likely to live in relatively more urban areas; therefore, our analysis might overestimate the effects of persecutions if the postwar expansion in education is negatively correlated with the fraction of the Jewish population in a given region in 1933.

Figure 1 shows that cohorts who were beyond the school-going age when the Nazi Party came into power and the younger birth cohorts who were too young to be affected from the persecutions have higher educational achievement if they were residing in regions where the percentage of the population that is Jewish in 1933 is higher. However, cohorts who were of school-age during persecutions attained similar levels of education both in high and low Jewish intensity regions. Therefore, the expulsion of the Jewish population has negatively affected the long-term academic performance of the school-aged children in regions with a high fraction of Jewish population. On the other hand, Figure 1 illustrates that regions with varying ex ante Jewish population exhibit parallel trends in educational attainment before the expulsions and after WWII, which further supports our identifying assumption. In addition, analysis presented in Figure 1 points out that regions with a high or a low percentage of Jews experienced a similar increase in educational attainment in the postwar era; therefore, our results are not confounded by the differential increase in educational attainment across urban and rural areas.

Moreover, we gauge the plausibility of the parallel trend assumption below by performing an additional falsification test. We repeat the analysis using the oldest cohorts who were already beyond school age before the expulsions and the younger cohorts who started their schooling after the postwar reconstruction period was over

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in the late 1950s. In this control experiment, we also find that the parallel trend assumption holds, which lends credence to the difference-in-differences estimation.

III. Data and Descriptive Statistics

The measure of Jewish fraction that we use for our main analysis is from Kessner (1935), who reports as his source the German population statistics from the German Population and Occupation Census conducted in 1933. Kessner (1935) provides city-level information on individuals affiliated with a particular religious community as a fraction of total population in 1933. The dismissal and emigration of German Jews began after the passage of the "Law for the Restoration of the Professional Civil Service" in 1933. The persecution of Jews was in principle based on a racial/ethnic criteria, not necessarily based on their religious affiliation (Evans 2005). With the aforementioned civil service law, the Nazi government required all employees and officials in the public sector to provide an "Aryan" certificate in order to identify the Jewish population (Heinemann 1999). After that, the Nuremberg Laws, which classifies racial Jews, was enacted in 1935 and expanded over time.[7] We use the percentage of the population that is affiliated with Jewish communities as the measure of Jewish presence in our study because no record of the number of racial Jews was recorded until 1939. Even though some of the Jews who were affected by these laws already emigrated, the 1939 German Census reports that 91.5 percent of the Jewish population also had religious affiliations (Blau 1950). Moreover, Mischlinge of the first and second degree constituted a large fraction of "racial" Jews who were not Jews by religion and their persecution started with the introduction of the Nuremberg Laws. So, the fraction of individuals who were Jews by religion was likely to be higher in 1933 than the 91.5 percent reported in 1939. Thus, we conclude that these data give accurate information on the fraction of Jewish population in Germany before the beginning of the persecutions.

We also assembled detailed information from various years of German Municipalities Statistical Yearbook on regional characteristics, including average income per capita in 1929, unemployment rate in 1932, the share of votes the Nazi Party (NSDAP) and the Communist Party (KPD) received in two federal elections in 1932, and area and population in 1933. Second, we collected region-year data on the number of teachers and professors before and after 1933 to assess the extent of dismissals among educators and more generally school inputs available to the affected cohorts. Finally, we compiled region-level information on the aggregate residential rubble in m^3 per capita in Germany by the end of WWII, which is what we use as a measure of wartime destruction.[8]

[7] In accordance with the provisions of the Nuremberg Laws, "racial" Jews are, (i) all persons with at least three grandparents who were racially full Jews. (ii) As Mischlinge ("crossbreed") of the first degree, all persons with two grandparents who were racially full Jews. (iii) As Mischlinge of the second degree, all persons with one grandparent who was racially a full Jew.

[8] Kaestner (1949) also provides information on the percentage of the residential dwellings destroyed in 1946 in the territory of former West Germany. The correlation between these two measures of WWII destruction is 0.9; thus, we present the results with rubble per capita measure.

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The data on individual and household characteristics are from the German Socio-Economic Panel (SOEP). SOEP is a household panel survey that is representative of the entire German population residing in private households. It provides a wide range of information on individual and household characteristics including parental background and childhood environment. SOEP also asks respondents whether they still reside in the city or area where they grew up. We utilize this question to ascertain whether individuals moved out of their childhood city or area at the time of the survey. We restrict the empirical analysis to individuals born between 1910 and 1960. These individuals are 25 and older in 1985; thus, it is very likely that they have completed their human capital formation. We dropped individuals born between 1928 and 1950 from the analysis since their human capital formation has been affected by exposure to WWII destruction and postwar reconstruction as shown in Ichino and Winter-Ebmer (2004).[9] In addition, our sample consists of German natives who were residing in former West Germany in 1985 including the Jewish Germans. We assume that the percentage of respondents who are Jewish is close to zero.[10]

We consider the effects of the Jewish expulsions at the smallest representative geographical units ("ROR" or "region") provided in SOEP.[11] The historical data on the ex ante percentage of Jewish population is available for 47 RORs. However, almost 85 percent of the German population at the time resided in these regions as they include all big cities and city states.[12] We obtain our historical dataset by digitalizing the city-level data on the fraction of the population that is Jewish and other historical variables from the 1930s German Municipalities Statistical Yearbooks. We then aggregate these variables according to the 1985 German regional (ROR) boundaries. This aggregation is possible since every city reported in the yearbooks belongs to only one region. We finally merge this aggregated ROR-level historical data with the 1985 wave of SOEP by an individual's ROR. We choose this year because this is the earliest date for which both households' ROR information and individual and parental characteristics are available. Furthermore, the 1985 wave of SOEP is only available for former West Germany.

Table 1 presents the descriptive statistics for the percentage of the population that is Jewish and regional characteristics before the Nazi Regime. Table 1 shows that, on average, 1.2 percent of the population was Jewish in former West Germany in

[9] Using the Andrews test, Ichino and Winter-Ebmer (2004) identify the potential German cohorts whose educational attainment was affected by WWII (i.e., the potential breaking points in the data). They show that educational attainment of cohorts born between 1928 and 1938 were affected by WWII. Therefore, we excluded these birth cohorts from our analyses in order to disentangle the true effect of the loss of Jewish population on the educational attainment of German children from potential WWII effects.

[10] Individuals' membership to religious institutions is reported in 2007 for the first time in SOEP. The tabulation of this variable shows that only 0.24 percent of respondents report that they are a member of other religious organizations. Other categories listed are Catholic, Evangelical, Other Catholic, Islamic religious organizations, and nondenomination. In addition, city-level data on the fraction of Jewish population residing in Germany in 1946 shows that only 0.15 percent of the population was Jewish in postwar Germany. Therefore, based on these data, we assume that the percentage of respondents in SOEP who are Jewish is close to zero.

[11] See Knies and Spiess (2007) for detailed information on RORs.

[12] These units are called Raumordnungsregionen (RORs) and are determined by the Federal Office for Building and Regional Planning (Bundesamt fuer Bauwesen und Raumordnung, BBR). West Germany has 75 different Raumordnungsregionen. RORs are "spatial districts" based on economic interlinkages and commuting flows of areas. RORs encompass the aggregation of Landkreise and kreisfreie Staedte (administrative districts, which are analogous to counties in the United States) and represent the center of the local labor market and surrounding small towns and rural areas (Jaeger et al. 2010).

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TABLE 1—DESCRIPTIVE STATISTICS FOR JEWISH POPULATION IN GERMANY

	RORs with above average Jewish population (1)	RORs with below average Jewish population (2)	All (3)	Difference standard error (difference) (4)
Percentage of Jews in 1933	1.957	0.715	1.189	1.242***
	(0.961)	(0.280)	(0.875)	(0.027)
Percentage of Jews in 1946	0.289	0.0727	0.155	0.217***
	(0.379)	(0.148)	(0.282)	(0.011)
Percentage of dismissed professors	14.45	6.512	10.74	7.935***
	(6.161)	(7.796)	(8.017)	(0.410)
Percentage change in the number of teachers	14.85	3.869	6.958	10.98***
	(7.428)	(10.43)	(10.86)	(0.519)
Area in km^2 in 1933	167.4	135.0	147.4	32.34***
	(82.74)	(141.2)	(123.2)	(5.144)
Population in 1932	555.4	338.3	421.2	217.1***
	(425.0)	(351.0)	(395.2)	(5.144)
Income per capita in 1929 (in RM)	382.8	350.0	362.1	32.80***
	(49.87)	(121.3)	(102.3)	(4.329)
Maximum number of observations	913	1,479	2,392	2,392

Notes: The means presented are weighted by population in the 1985 SOEP. Standard deviations are in parentheses.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

1933.[13] However, there was a sizable variation across regions in the density of Jewish population. For example, almost 4 percent of the population in Berlin and Frankfurt was Jewish. In contrast, the Jews constituted only 0.4 percent of the population in Bremen and Essen. Table 1 also highlights that regions where a higher fraction of the population is Jewish are larger in area and have a higher ex ante population and average income per capita as suggested by the historical records. This underlines the fallacy of relying only on cross-regional variation in the percentage of the population that is Jewish to identify the long-term educational effects of expulsions. Therefore, the difference-in-differences strategy we propose uses within-region, cross-cohort variation to identify the effects of purges and controls for differences between birth cohorts that are common across German regions. The observed differences in regional population density and per capita income may potentially suggest differences in ex ante and ex post cohort-specific trends in children's educational outcomes that are unrelated to the dismissals. In Section IV, we assess whether there are differential pre- and post-cohort-specific trends by performing a falsification test and find no evidence for such trends.

[13]The percentage of the population that is Jewish is slightly higher in Table 1 than the aforementioned average of 0.8 because the 1935 German Municipalities Statistical Yearbook reports the percentage of Jewish population for cities with 50,000 and more inhabitants. Hence, Jews were more likely to live in urban areas; data reported in the 1935 German Municipalities Statistical Yearbook overestimate the percentage of Jews. However, we identify the long-term direct and spillover effects of the Jewish dismissals on German children's human capital formation using across-cohort within-region variation in the percentage of Jewish population; therefore, our results are not affected by this.

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TABLE 2—DESCRIPTIVE STATISTICS, SOEP DATA

	RORs with above average Jewish population (1)	RORs with below average Jewish population (2)	All (3)	Difference standard error (difference) (4)
Years of schooling	11.27	11.08	11.15	0.19
	(2.582)	(2.445)	(2.499)	(0.11)
Has basic education	59.36	58.76	58.99	0.61
(*Hauptschule*)	(49.14)	(49.24)	(49.20)	(2.07)
Has more than basic education	16.21	15.89	16.01	0.32
	(36.87)	(36.57)	(36.68)	(1.54)
Mother with basic education	87.28	89.22	88.49	−1.94
	(33.35)	(31.03)	(31.93)	(1.45)
Father with basic education	81.53	83.36	82.67	−1.83
	(38.83)	(37.26)	(37.86)	(1.73)
Age in 1985	47.30	47.29	47.29	0.010
	(18.35)	(18.54)	(18.46)	(0.78)
Female	53.45	53.82	53.68	−0.37
	(49.91)	(49.87)	(49.87)	(2.10)
Rural	41.03	39.09	39.83	1.94
	(49.22)	(48.81)	(48.97)	(2.06)
Maximum number of observations	913	1,479	2,392	2,392

Notes: Data are from the 1985 SOEP. The sample consists of individuals born between 1910 and 1960. Individuals born between 1928 and 1950 are dropped from the analysis since they were exposed to WWII destruction and post-war reconstruction. Standard deviations are presented in parentheses. All variables except years of schooling and age are presented in percentages.

Table 2 shows the descriptive statistics for the outcomes and the main individual-level control variables we use in our estimation. The main educational outcome we focus on is the completed years of schooling. SOEP asks respondents about their highest educational attainment; then in the data files it generates the individual's years of schooling using these attainment categories. While most of our regression analysis is with the years-of-schooling measure, we also present results using the highest educational attainment categories. These educational outcomes are measured five decades after the persecutions and reflect the outcomes of German-natives who lived to 1985 or later.

IV. Estimation Results

Table 3 reports the results of estimating equation (1) where the dependent variable is completed years of schooling. Each column is from a separate regression that controls for region and birth year fixed effects along with female and rural dummies. The difference-in-differences estimate, β, is reported in the first row. It is negative and significant at the 99 percent level of confidence in every specification. We present the difference-in-differences estimate for the entire population in column 1 of Table 3. β is -0.43 in column 1 of Table 3, suggesting that school-aged children and young adults in a region with average percentage of ex ante Jewish population attain 0.5 fewer years of schooling. This is the difference-in-differences coefficient β (-0.43) multiplied by the percentage of the population that is Jewish in 1933

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TABLE 3—EFFECTS OF JEWISH EXPULSIONS ON YEARS OF SCHOOLING

	(1)	(2)	(3)	(4)
Percentage of Jews in 1933 × born 1910–1927	−0.4292***	−0.3460***	−0.3241***	−0.3125***
	(0.0742)	(0.0748)	(0.0725)	(0.0698)
Percentage of Jews in 1933 × born 1910–1927 × mother has more than basic education		0.1790		−0.3945
		(0.2628)		(0.3914)
Percentage of Jews in 1933 × born 1910–1927 × father has more than basic education			0.5379*	0.7227**
			(0.2678)	(0.3264)
R^2	0.216	0.304	0.331	0.352
Observations	2,385	2,061	2,037	2,006

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1951 and 1960. Each column is from a separate regression where the main treatment effect varies by parental education in columns 2–4. Each column controls for region and birth year fixed effects, gender, and rural dummies. Columns 2–4 also control for the main effects of parental human capital.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

(1.19) reported in Table 1. We can also interpret the magnitude of β by comparing the educational attainment of school-aged children who were in Frankfurt (3.25 percent of its population was Jewish) to children who were in Bremen (0.4 percent of its population was Jewish) during the Nazi Regime. Using this comparison, column 1 of Table 3 suggests that children in Frankfurt had 1.2 fewer years of schooling compared to children in Bremen as a result of the Jewish expulsions. It is likely that the loss of Jewish professionals had adverse effects on the quality of education in addition to the quantity of education. Therefore, the analysis presented in Table 3 provides a lower bound estimate for the potential negative consequences of the purge of Jewish population on children's long-term academic achievement.[14]

Columns 2–4 of Table 3 present the results incorporating family background characteristics, such as father's and mother's educational attainment, which may also serve as a proxy for parents' economic well-being. The first row in these columns reports the estimation results for children whose parents had a basic school degree (*Hauptschule*) or less.[15] Results summarized in columns 2–4 of Table 3 reveal that children with less educated parents had a greater reduction in their educational attainment (first row). The point estimates of the interaction terms suggest that the negative effect of expulsions is mitigated for children whose parents have more than basic education (second and third rows). This differential effect may work literally through parental education (e.g., more educated parents value education more and so ensure their children are educated too even if negative shocks occur) or through other channels correlated with parental education such as family income or wealth (e.g., wealthy families can afford to educate their children and can hire private tutors or send children to boarding schools when necessary).

[14] As robustness, we estimate the same specifications as in Table 3 using the decline in the percentage of the population that is Jewish between 1933 and 1946 as a measure of loss of Jewish population. We find quantitatively similar results with this alternative measure. These analyses are available upon request.

[15] The basic school diploma (*Hauptschule*) is granted after nine years of schooling in Germany. As shown in Table 2, the majority of children have parents with basic education or less (83 percent of fathers and 89 percent of mothers in our sample completed basic education or less).

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Table 3 shows that the reduction in the percentage of the population that is Jewish led school-aged children to attain approximately 0.5 fewer years of schooling. However, young adults and children may have been differentially affected from the persecution of the Jewish population. For instance, at the college level, the persecution of Jewish professors could decrease the demand for college education among students. On the other hand, young Jewish adults may have a higher college enrollment rate than the rest of the population before expulsions. Historical sources report that the Jewish students constituted 3 to 4 percent of the university students before the Nazi government banned them from entering into universities as a part of their systematic persecution policy (Evans 2005, 299). Therefore, the effect on admission to competitive colleges among non-Jewish students is unclear, a priori. For children at the lower and upper secondary school, the decline in the availability of teachers may have a larger effect since there is no competition in admissions to these schools. Moreover, the purge of Jewish professionals may lead to less competition in the labor market and more uncertainty for the future, thereby increasing the number of early dropouts from these schools. For primary school-aged children, the effects of dismissals were likely smaller compared to other groups. These children were at the beginning of their educational career; therefore, they may have enough time to recover from the adverse effects of expulsions. In addition, these children were affected by the persecutions for a shorter period of time than the children who were at the lower and upper secondary school age during the Nazi Regime. To assess level of education at which the adverse effect of expulsions is more profound, we estimate the following difference-in-differences specification using OLS:

$$(2)\quad Y_{irtm} = \alpha + \beta_m(JewishFr_r \times AffectedCoh_{it}) + \delta_r + \gamma_t + \pi'\mathbf{X}_{irtm} + \epsilon_{irtm},$$

where the outcome of interest, Y_{irtm} is a dummy variable that indicates whether the individual i born in year t, in region r, completed m years of schooling or more. β_m, for $m = 9$ to 18, is the estimated effect of the persecution of the Jews on the probability of completing each level of education. Each point in Figure 2 is from a separate regression where the outcome is a dummy variable that takes a value of 1 if an individual completed m years of schooling or more. We find in Figure 2 that the purge of Jewish professionals disrupts school-aged children's educational attainment in all levels of education; the point estimates are always negative. Figure 2 also reports that the affected cohorts in a region with an average percentage of Jewish population (1.19) are 3 percentage points less likely to finish basic high school (9 years of schooling) and 5–6 percentage points less likely to complete technical high school or Gymnasium (12–13 years of schooling). Moreover, these cohorts are 3 percentage points less likely to have a college degree (16–17 years of schooling) and a graduate degree (18 years of schooling) because of the dismissal of Jewish professionals.

We provide additional analysis on the cohort-specific effects of Jewish expulsions in Appendix Table 2. For this analysis, we separately estimate the effects of dismissals on the cohorts who were only of college age in 1933, many of whom already completed their education decisions (19–23 years of age in 1933), children who were of primary and secondary school age in 1933 and were more likely to

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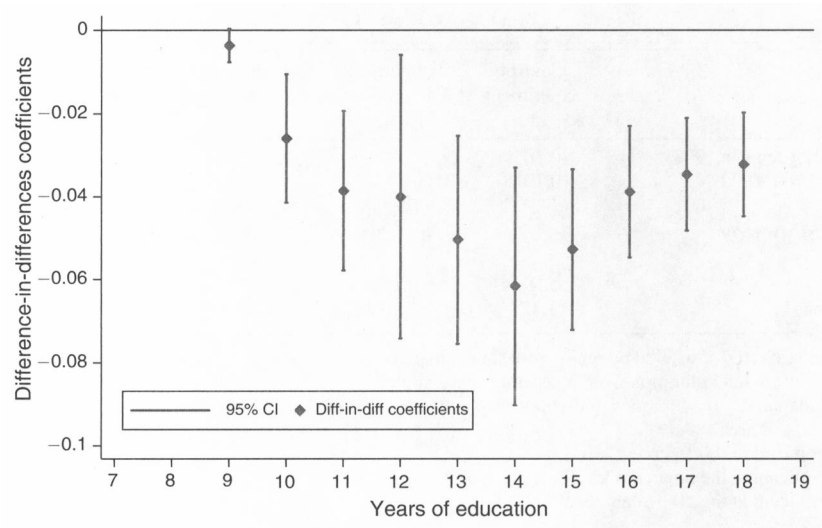

FIGURE 2. ESTIMATED EFFECT OF JEWISH EXPULSIONS ON FULL DISTRIBUTION OF EDUCATION

Note: Each point in this figure is the difference-in-differences estimate from a separate regression where the outcome is a dummy variable that takes a value of 1 if an individual completed *m* years of schooling or more and zero otherwise.

be affected by expulsion (6–18 years of age) and children who were younger than 6 in 1933 and had time to recover from the adverse effects of persecutions. Taken together, results presented in Appendix Table 2 suggest that cohorts enrolled in primary and secondary school when the Nazi Party came into power were affected more on average than those who were in college or who had not yet entered school.

A. *Robustness Checks and Potential Confounding Factors*

Results presented in Table 3 and Figure 2 rest on the parallel trend assumption, which assumes that the affected and the control cohorts would have had parallel trends in educational attainment in the absence of the Jewish persecutions. That is, the coefficient for interaction between dummy for being born 1910–1927 and the fraction of ex ante Jewish population in each region would be zero in the absence of the expulsions. We first quantify the validity of the parallel trend assumption in Figure 1. As aforementioned in the identification strategy section, Figure 1 illustrates that regions with varying intensity of Jewish population exhibit parallel trends in educational attainment before the expulsions and after WWII. Therefore, Figure 1 suggests that our results are not confounded by the differential ex ante or ex post cohort-specific trends across regions.

We provide further evidence on the parallel trend assumption by performing an additional falsification test. For this control experiment, we focus on the oldest cohorts who completed their education before the Nazi Party came into power in 1933 and the youngest cohorts who began their human capital formation after the postwar reconstruction period was over in the late 1950s. The oldest cohorts (i.e., those born between 1900 and 1909) are treated as the "placebo" affected cohorts

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TABLE 4—VALIDITY CHECKS

	Control experiment (1)	Internal migration (2)	Nonmovers only (3)	Mortality (4)	Cohort size (5)
Percentage of Jews in 1933 × born 1900–1909	−0.1059				
	(0.1018)				
Percentage of Jews in 1933 × born 1910–1927		−0.0240	−0.3853***	−0.0044	−0.1226
		(0.0183)	(0.1396)	(0.0095)	(0.2089)
R^2	0.189	0.111	0.226	0.521	0.511
Observations	1,479	2,379	1,303	2,385	2,385

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1951 and 1960. Each column is from a separate regression. Each column controls for region and birth year fixed effects, gender, and rural dummies. Individuals are coded as movers if they report that they no longer reside in their childhood city or area.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

and the youngest cohorts who were born between 1951 and 1960 are treated as the "placebo" control cohorts (this is also our original control group). Results are presented in column 1 of Table 4. We find that differences in educational achievement between the oldest and the youngest cohorts are similar across regions. Moreover, this analysis suggests that the educational attainment of our control group has not been affected by the persecution of the Jewish population. Thus, these results lend credence to the identification assumption in equation (1).

Another potential confounding factor for results summarized in Table 3 and Figure 2 is the probability of the nonrandom internal migration across regions. It is well documented, however, that Germany has historically low levels of geographic mobility, and that mobility is particularly low during childhood and early adulthood (Rainer and Siedler 2009, Pischke and von Wachter 2005, and Hochstadt 1999). Therefore, internal migration should yield at most minor consequences for our estimates. Nonetheless, as robustness, we test whether individuals' migration decision is associated with the percentage of the population that is Jewish in each region in 1933 using the probability of moving as the dependent variable. Individuals are coded as movers if they report that they no longer reside in their childhood city or area in 1985. The affected and the control groups for this specification are the same as in Table 3. The difference-in-differences estimate for probability of moving is close to zero and statistically insignificant as summarized in column 2 of Table 4. This finding bolsters our confidence that individuals did not choose their final destination according to the fraction of the Jewish population of the region.

We provide further evidence on the lack of systematic internal migration by restricting our analysis to individuals who still live in the city or area where they grew up. In column 3 of Table 4, we find that the difference-in-differences estimate for nonmovers is very similar to the estimates for the entire population (the difference-in-differences estimates for the entire population and nonmovers lie within each other's 95 percent confidence intervals.) These results further support our previous findings and indicate that nonrandom internal migration is unlikely to be a concern for the results presented in Table 3 and Figure 2.

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Another potential concern is the probability of differential mortality across regions with varying intensity of ex ante Jewish population. For this analysis, we take advantage of the panel structure of SOEP, which enables us to analyze the mortality of the respondents between 1985 and 2011. The mortality variable refers to a dummy variable that takes a value of 1 if an individual has a recorded death year sometime between 1985 (the beginning of our sample) and 2011, and zero otherwise.[16] The difference-in-differences estimate for mortality in column 4 of Table 4 is close to zero and insignificant; therefore, our analysis is not confounded by the differential mortality rates across regions.

We provide an additional check for composition bias by estimating equation (1) using the size of each annual cohort in 1985 as an outcome. The results appear in column 5 of Table 4. This specification provides a formal test of whether individuals born in certain birth years are overrepresented or underrepresented in 1985 in regions with varying intensity of ex ante Jewish population. The estimation results show that there is no statistically significant variation across different birth cohorts in 1985 in different regions. Therefore, we find that the estimated impacts of Jewish dismissals on children's human capital investment are not an artifact of differential survival rates across regions.

Finally, analyses presented in Appendix Table 3 bolster our confidence that our results are robust to the choice of affected cohorts. In Appendix Table 3, we estimate equation (1) where we define the affected cohorts using different birth year cutoffs based on Appendix Table 1. In column 1 of Appendix Table 3, the affected cohorts encompass individuals born between 1910 and 1923. This is the most conservative definition of the affected group; hence, these cohorts were beyond compulsory schooling age on the onset of WWII so their schooling has not been affected by WWII destruction. In column 2 of Appendix Table 3, we present the difference-in-differences estimate with the affected cohorts used in the main analysis in Table 3. In column 3 of Appendix Table 3, we extend the affected cohorts to individuals born between 1910 and 1933. These cohorts were six and older on the onset of WWII in 1939; therefore, their education might have been potentially affected from the persecution of Jewish population until 1939. The last column presents the difference-in-differences estimate, where individuals born between 1910 and 1938 constitute the affected group. This is the most inclusive definition of the affected group, where at least one year of the educational attainment of these cohorts had been potentially interrupted by the loss of Jewish population. Individuals born in 1910 may have been in the last year of their college education when the dismissals started in 1933. On the other hand, individuals born in 1938 were in the first year of the primary school just before the end of the Nazi Regime in 1945. Thus, they are potentially the youngest cohorts that have been affected by the persecution of Jewish population in Nazi Germany. We find quantitatively similar results to the baseline specification in all columns in Appendix Table 3. These analyses lend credence to our estimation results and suggest that our results remain statistically similar with a different categorization of the affected group.[17]

[16] Information on an individual's death year in SOEP comes from official vitality records.

[17] We find nearly identical results when we extend the college finishing age to 25 years of age instead of 23.

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B. *Potential Historical Confounding Events*

During the time period we focus on in our paper, Germany also experienced other political and economic events, including World War II. Moreover, compulsory schooling laws changed in German states starting from the late 1940s. The nationwide effects of these events are captured by the birth year fixed effects in our analysis. We also use a lower level of geographical aggregation than state in our estimation, allowing us to account for potential state-specific policies. However, it is possible that regions with a higher percentage of Jewish population may have been differentially impacted by these events, which may raise potential concerns on the interpretation of our analyses. In this subsection, we formally test whether our results are confounded by these events; results are presented in Table 5. In all columns in Table 5, we control for state-cohort trends in addition to region and birth year fixed effects and female and rural dummies. The first row in Table 5 reports the difference-in-differences estimate for the percentage of ex ante Jewish population.

One of the potential concerns for the interpretation of our results is the state-specific policies, which might differentially influence the affected cohorts in regions with a higher fraction of ex ante Jewish population. The extent of such potential bias is largely mitigated because we use a lower level of geographical aggregation than state in our analysis to allow us to explore within-state variation. Moreover, we formally test whether our results are sensitive to the inclusion of state-cohort trends. We find a quantitatively similar difference-in-differences estimate, -0.436 with standard error of 0.070. This suggests that our results are not confounded by state-specific policies.

In addition, the compulsory schooling has increased from eight to nine years in Germany after the late 1940s. The timing of the reform was determined by state governments and varied at the state level. To address this potential concern, we first adjust the individuals' generated years of schooling variable reported in SOEP. Following Pischke and von Wachter (2005) and Cygan-Rehm and Maeder (2013), we replace the years of schooling of individuals who completed *Hauptschule* before the introduction of the reform with eight years. We use this adjusted years of schooling measure for all of our empirical analysis. Second, we control for the interaction between state dummies and birth year dummies for individuals who were affected by the compulsory schooling law in column 1 of Table 5. The difference-in-differences estimate in column 1 of Table 5 remains statistically and quantitatively similar to the baseline specification, contributing to our confidence that our results are robust to the timing of the compulsory schooling reform.

The WWII destruction experienced by the region also might have been associated with the percentage of the Jewish population residing in this region during the Nazi Regime. To address whether school-aged children had been differentially affected by WWII destruction, we estimate the baseline specification controlling for the interaction between regional wartime destruction measure (rubble in m^3 per capita) and an indicator for being in the affected group. Column 2 of Table 5 shows that the difference-in-differences estimate for WWII destruction is very small and statistically insignificant, suggesting that there was no variation in exposure to wartime destruction across regions of varying intensity of ex ante Jewish population.

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TABLE 5—HISTORICAL CONTROLS

	Compulsory schooling (1)	WWII (2)	Refugees (3)	Unemp. rate (4)	Elections (5)	Population income (6)	Urban share (7)	All (8)
Percentage of Jews × born 1910–1927	−0.270**	−0.421***	−0.372***	−0.435***	−0.445***	−0.422***	−0.455***	−0.319**
	(0.120)	(0.068)	(0.090)	(0.070)	(0.068)	(0.063)	(0.067)	(0.140)
Rubble per capita × born 1910–1927		−0.011						−0.005
		(0.015)						(0.018)
Percentage of refugees × born 1910–1927			0.023					0.034
			(0.018)					(0.028)
Unemployment rate × born 1910–1927				0.035				0.072
				(0.046)				(0.083)
Percentage of votes to Nazi Party × born 1910–1927					0.013			0.001
					(0.016)			(0.020)
Percentage of votes to Communist Party × born 1910–1927					−0.014			−0.008
					(0.016)			(0.020)
Population size × born 1910–1927						−0.000		−0.000
						(0.000)		(0.000)
Income per capita × born 1910–1927						−0.001		−0.001
						(0.001)		(0.001)
Urban share × born 1910–1927							−0.005	−0.002
							(0.006)	(0.010)
R^2	0.223	0.221	0.221	0.221	0.221	0.224	0.221	0.226
Observations	2,385	2,356	2,385	2,385	2,385	2,335	2,385	2,306

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1951 and 1960. Each column controls for region and year of birth fixed effects, gender, rural dummies, and state-cohort trends. Data on unemployment rate, percentage of votes to the Nazi Party and the Communist Party, and population size are from 1932. We also control for the percentage of refugees in 1961 and income per capita in 1929.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

A related concern is refugees or people who fled to Germany from former parts of Germany and Soviet Zone/GDR after WWII.[18] These refugees might have settled in regions with a higher percentage of ex ante Jewish population to fill the positions vacated by Jewish professionals. As an attempt to address this potential concern, we use the official 1961 regional-level refugee data provided by Redding and Sturm (2008). This is the earliest data we are able to access on refugees that serves as a proxy for percentage of refugees residing in German regions by the end of WWII. We include this measure interacted with being in the affected cohort as a control variable into our baseline specification; results are summarized in column 3 of Table 5. We find no evidence for differential impacts of refugees.

In column 4 of Table 5, we consider the macroeconomic conditions German regions were in during the early 1930s. The macroeconomic conditions and the employment opportunities in general might have had differential impacts in regions with a higher percentage of Jews, altering the incentives for human capital investment among German children in these regions. In a high unemployment

[18] See Burchardi and Hassan (2013) for a discussion of the settlement of refugees after WWII in Germany.

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environment, these children might have chosen to work in lower skilled jobs instead of investing in higher education. Alternatively, adverse labor market conditions may have induced children to stay at school longer and acquire more education. As an attempt to address this potential concern, in column 4 of Table 5, we use the regional unemployment rate in 1932 as a proxy for the macroeconomic conditions in German regions. The difference-in-differences estimate for the percentage of the Jewish population in 1933 remains virtually unchanged when we control for the interaction between the unemployment rate in 1932 and an indicator for being in the affected group. Therefore, children's educational attainment was not differentially affected by differences in the regional unemployment rate.

An additional concern is the differential Nazi Party and the Communist Party support across German regions. Regions with a higher fraction of ex ante Jewish population might have had a stronger support for the Nazi Party and Anti-Semitic policies implemented during the Nazi Regime. Moreover, members of the Communist Party were also expelled from civil service in Germany with the civil service law. Deichmann (2001) and Waldinger (2010) suggest that only a small fraction of the professionals were dismissed because of being "politically unreliable"; thus, it is unlikely that our results are confounded by differential Communist Party support across regions. Nevertheless, as robustness, we assembled regional-level data on the percentage of votes the Nazi Party and the Communist Party of Germany received in the federal elections in 1932 to address this potential concern. Using the federal election of November 1932 results, we control for the interaction between the percentage of votes received by the Nazi Party and the Communist Party and being in the affected cohort. Estimation results summarized in column 5 of Table 5 mimic our previous findings.[19]

Nazi policies and potentially other policies might have differentially affected individuals in larger and more prosperous regions. In addition, RORs may differ in terms of the urban population they encompass. In column 6 of Table 5, we control for the interaction of the regional population in 1932 and income per capita in 1929 with an indicator for being in the affected group. Similarly, in column 7 of Table 5, we include the interaction between the percentage of urban population in each region and being in the affected cohort to account for potential difference in urbanization across regions. The difference-in-differences estimates in columns 6 and 7 of Table 5 reveal that there is no variation in children's educational attainment across regions of varying income per capita, population, and urbanization.

In the last column of Table 5, we control for all of the potential confounding factors. The difference-in-differences estimate for the percentage of ex ante Jewish population is still economically and statistically significant in this specification. Therefore, analyses presented in Table 5 suggest that the dismissal of Jewish professionals had detrimental long-term effects on German children's educational attainment even after we account for state-specific policies, regional macroeconomic conditions, the Nazi and the Communist Party support, WWII, refugees, the compulsory schooling reform, and urbanization.

[19] We find quantitatively similar results when we use the results from the federal election that took place in July 1932.

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TABLE 6—HETEROGENEITY IN THE EFFECTS OF JEWISH EXPULSION ON YEARS OF SCHOOLING

	Female only (1)	Male only (2)	Urban only (3)	Rural only (4)	Top quartile Jewish population (5)	Number of years affected (6)
Percentage of Jews in 1933 × born 1910–1927	−0.3548***	−0.5188***	−0.4527***	−0.3882***	−0.6752*	
	(0.1130)	(0.1386)	(0.1049)	(0.1098)	(0.3575)	
Percentage of Jews in 1933 × number of years affected						−0.0316***
						(0.0056)
R^2	0.275	0.135	0.184	0.305	0.215	0.203
Observations	1,277	1,108	1,435	950	2,385	3,509

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1951 and 1960. In column 5, "Top Quartile Destruction" is a dummy variable which takes a value of 1 for interaction of being in regions with highest Jewish fraction during the Nazi Regime and being born between 1910 and 1927. Each column controls for region and birth year fixed effects, gender, and rural dummies.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

C. *Heterogeneity and Channels*

In this subsection, we provide formal evidence on the heterogeneity and the potential mechanisms through which persecution of the Jews may have affected school-aged children. The results are reported in Table 6. Women constitute 56 percent of our affected cohorts. Therefore, it is of interest to test whether our results are similar across gender. The first and second columns of Table 6 report the difference-in-differences estimates for only female and male samples, respectively. Findings summarized in these columns show that the difference-in-differences estimate is larger in magnitude for the male sample; however it is not statistically different than the difference-in-differences estimate for the female sample. These gender-specific analyses also suggest that the main education results presented in Table 3 are not an artifact of the higher number of women in the 1985 SOEP. Columns 3 and 4 of Table 6 consider whether the long-term effects of purges differ by an individual's urban status. We find that the adverse educational effects of persecutions are statistically similar for the affected cohorts in urban and rural areas. Therefore, our results are not driven by the urban areas.

Additionally, one may expect the effects of the persecutions to be nonlinear, e.g., if a fraction of ex ante Jewish population surpasses a certain level, then the detrimental effects become especially large, otherwise the effects are modest or negligible. To explore whether the negative effects of expulsions are more pronounced in regions with the highest percentage of ex ante Jewish population, we divide the Jewish population intensity measure into quartiles. In this specification, we control for the interaction of these quartiles with being in the affected cohort. We find that the adverse effects of the persecutions are larger in regions that lost a higher fraction of their population due to the expulsions. Column 5 of Table 6 shows that children in

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the top quartile attain 0.7 fewer years schooling relative to the control cohorts; this effect is almost twice the estimate presented in column 1 of Table 3.[20]

In column 6 of Table 6, we replace the affected group dummy with a continuous measure of the number of school-age years an individual was potentially affected by the persecutions.[21] We find that children in a region with an average fraction of ex ante Jewish population attained 0.44 fewer years of schooling if they were potentially affected by the purge of the Jewish population during the entire duration of the Nazi Regime. This is the difference-in-differences coefficient $\beta\,(-0.032)$ multiplied by the average population-weighted Jewish population in 1933 (1.19 percent) in Table 1 and the duration of the Nazi Regime (12 years). This additional analysis shows that the estimation results presented in Table 3 also hold when the affected cohort dummy is replaced with a continuous measure of the number of school-age years a cohort was exposed to persecutions.

Having shown that school-aged children have lower educational attainment due to the systematic persecution of the Jewish population, it is of interest to explore the channels underlying this causal association. For example, the dismissal of school principals, teachers, and professors might have adversely affected the educational decisions of German children. Similarly, declining teacher quality and bigger class sizes could have increased the number of early dropouts. Another potential channel is the change in the peers in school. If the Jewish students were on average stronger students, the dismissal and exile of Jews could have negative effects on the non-Jewish students' academic performance. Moreover, the dismissal of Jewish population from all civil service jobs might have provided new employment opportunities for the rest of the German population, thereby changing the incentives for human capital investment among non-Jewish children and young adults. Finally, the persecutions might have caused psychological effects on German children due to social unrest and violence in the streets, the loss of peers and friends and uncertainty about the future.

Figure 3 illustrates the association between the percentage of dismissed science professors and the percentage of the population that is Jewish in each region in 1933.[22] Similarly, Figure 4 displays the relationship between the percentage change in the number of school teachers between 1932 and 1934 and the percentage of Jewish population in a given region in 1933. Hence, data on the exact number of dismissed teachers in each region is not available; we use the change in the number of teachers between 1932 and 1934 in each region, which was mainly driven by dismissals, as a proxy for the dismissed teachers. These figures show that regions with a higher fraction of ex ante Jewish population had a greater decline in the percentage of science professors and experienced a greater percentage change in the number of teachers after the Nazi Party seized power in 1933.

[20] The point estimate for the interaction of being in the third quartile and the affected cohort is -0.3152 with a standard error of 0.2493, while the corresponding coefficient for the second quartile is -0.1754 with a standard error of 0.2717.

[21] We utilize the last column of Appendix Table 1, which reports the total years of exposure to generate this variable.

[22] For the percentage of dismissed science professors, we use data presented in Table 2 in Waldinger (2012).

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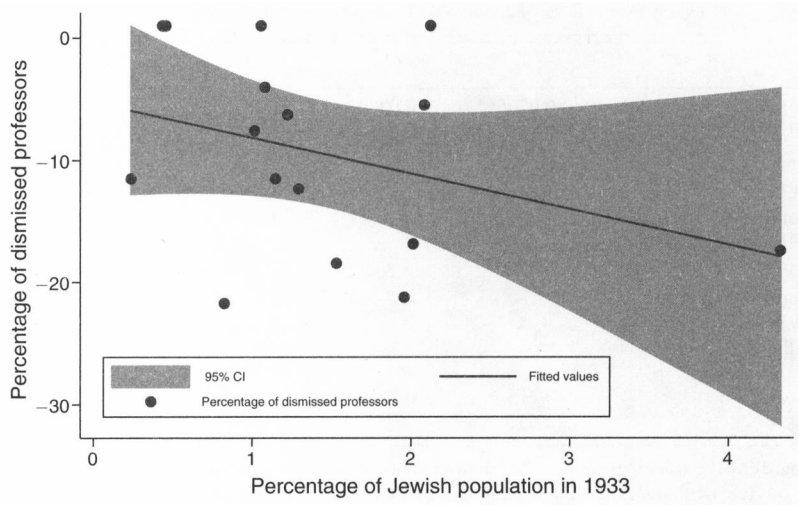

FIGURE 3. PERCENTAGE OF DISMISSED SCIENCE PROFESSORS AND THE PERCENTAGE OF JEWISH POPULATION IN GERMAN RORs

Source: Waldinger (2012)

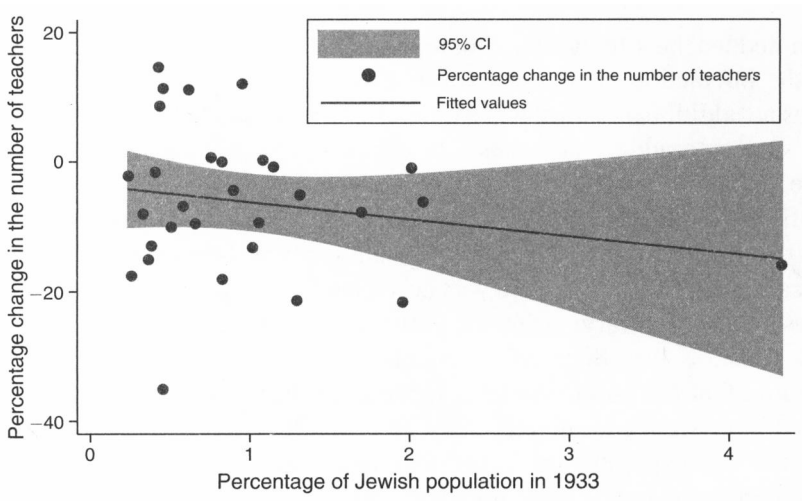

FIGURE 4. PERCENTAGE CHANGE IN THE NUMBER OF TEACHERS AND THE PERCENTAGE OF JEWISH POPULATION IN GERMAN RORs

Source: 1930s German Municipalities Statistical Yearbook

In Table 7, we provide more formal evidence on the effects of dismissed teachers on German children's educational attainment. Consistent information on the number of teachers in 1932 and 1934 is only available for the subsample of our data; therefore, results presented in Table 7 are only restricted to these regions. Analysis in column 1 of Table 7 suggests that school-aged children in regions with an average percentage of the dismissed teachers attained 0.2 years less schooling. This is the difference-in-differences estimate (0.027) multiplied by the average percentage of

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Table 7—Effects of Dismissed Teachers on Years of Schooling

	(1)	(2)	(3)	(4)	(5)
Percentage of dismissed teachers × born 1910–1927	−0.0265**	−0.0223*	−0.0254**	−0.0274**	−0.0274**
	(0.0122)	(0.0122)	(0.0124)	(0.0123)	(0.0121)
WWII destruction × born 1910–1927		−0.0241	−0.0132	−0.0064	−0.0072
		(0.0203)	(0.0175)	(0.0171)	(0.0161)
Percentage of dismissed teachers × born 1910–1927 × mother has more than basic education			0.0374		0.0094
			(0.0322)		(0.0465)
Percentage of dismissed teachers × born 1910–1927 × father has more than basic education				0.0349	0.0374
				(0.0243)	(0.0321)
R^2	0.213	0.214	0.298	0.335	0.351
Observations	1,713	1,713	1,480	1,464	1,440

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1951 and 1960. Each column is from a separate regression where main treatment effect varies by parental education in columns 3–5. Each column controls for region and birth year fixed effects, gender, and rural dummies. Columns 3–5 also control for main effects of parental human capital.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

the dismissed teachers reported in column 3 of Table 1. In column 2 of Table 7, we include the interaction of regional wartime destruction with being in the affected cohorts as an additional control to account for the potential WWII effects. This specification serves an additional robustness check since we exclude the birth cohorts that were shown to be affected from WWII in Ichino and Winter-Ebmer (2004) to capture the true effects of dismissals. Similar to the analysis presented in Table 5, we find that the difference-in-differences estimate for WWII destruction is statistically insignificant, suggesting that there was no variation in exposure to wartime destruction across regions of varying intensity of the dismissed teachers. In the last three columns, we allow the effects of the dismissed teachers to vary by parental education. We find that the point estimates for teacher dismissals are smaller in magnitude for children from educated families; however, they are not statistically different from the rest of the population. Taken together, analyses presented in Table 7 suggest that the loss of Jewish educators had substantial, enduring effects on German children's academic success.

In Appendix Table 5, we further explore the possibility of using the percentage of Jewish population in 1933 as an instrument for the percentage change in the number of teachers. Similar to Table 7, we only focus on a subsample of regions in this analysis where data on the number of teachers in 1932 and 1934 are available. The lower panel in Appendix Table 5 shows that first-stage estimates are statistically significant at a 1 percent significance level. This finding is consistent with the evidence presented in Figure 4, which suggests that regions with a higher fraction of ex ante Jewish population experienced a sizable decrease in the number of teachers due to persecutions. In the upper panel of Appendix Table 5, the two-stage least squares estimates are presented. The 2SLS estimate in column 1 of Appendix Table 5 suggests that a percentage decrease in the number of teachers is associated

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with 0.15 fewer years of schooling among the affected cohorts. The validity of the 2SLS estimates, however, rely on the satisfaction of exclusion restriction. The exclusion restriction assumes that the percentage of the Jewish population in 1933 affects the academic success of German children only through the dismissal of teachers. However, with the Nazi's civil service law, in addition to Jewish educators, Jews practicing other white-collar professions and those working in any type of civil service job were also expelled from their posts. Our study, therefore, estimates not only the direct effects of the persecutions caused by the dismissal of educators but also the potential spillover effects arising from a general loss of human capital due to the systematic expulsion of the Jewish population. Thus, the 2SLS analyses presented in Appendix Table 5 warrant caution.

V. Conclusion

In this paper, we estimate the long-term direct and external effects of the systematic persecution of the Jewish population in Nazi Germany. Using region-by-cohort variation in the percentage of the population that is Jewish in 1933, we find that children who were school-aged during the persecutions completed fewer years of schooling and had a lower probability of finishing a technical and academic high school, or university. The reduction in educational attainment is borne disproportionately by girls and boys living in regions with the highest fraction of Jewish population and whose parents were less educated. We also find that the dismissal and exile of Jewish science professors and teachers significantly contributed to the estimated long-term educational effects of the persecutions, while loss of peers and friends, uncertainty about the future, unrest in the streets and changes in the labor market are other potential mechanisms explaining our results.

Findings in this paper may shed light on the potential long-term legacies of large-scale human capital loss that could be caused by the economic and the political turmoil experienced in many countries around the globe. In 2012 alone, there were 32 active armed conflicts in 26 locations (Themnér and Wallensteen 2014). With the Arab Spring, many countries in the Middle East were torn by the political instability and violence, professionals supporting the old regimes were dismissed from their jobs after recent transitions. Our findings show that regions that lost a large number of highly skilled professionals not only experience deterioration in macroeconomic outcomes as shown in the previous literature, but such large-scale human capital loss has more substantial and enduring consequences for children and young adults. Hence, the detrimental effects of the Jewish persecutions in Nazi Germany are still present five decades after the Holocaust; our findings underline the potential importance of policies primarily targeting school-aged children following these large-scale negative human capital shocks.

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Appendix

Table A1—Affected Cohorts

Birth cohort	Years affected in primary school (grades 1–4)	Years affected in lower secondary (grades 5–10)	Years affected in upper secondary (grades 11–13)	Years affected in university	Total
1909	0	0	0	0	0
1910	0	0	0	1	1
1911	0	0	0	2	2
1912	0	0	0	3	3
1913	0	0	0	4	4
1914	0	0	0	5	5
1915	0	0	1	5	6
1916	0	0	2	5	7
1917	0	0	3	5	8
1918	0	1	3	5	9
1919	0	2	3	5	10
1920	0	3	3	5	11
1921	0	4	3	5	12
1922	0	5	3	4	12
1923	0	6	3	3	12
1924	1	6	3	2	12
1925	2	6	3	1	12
1926	3	6	3	0	12
1927	4	6	2	0	12
1928	4	6	1	0	11
1929	4	6	0	0	10
1930	4	5	0	0	9
1931	4	4	0	0	8
1932	4	3	0	0	7
1933	4	2	0	0	6
1934	4	1	0	0	5
1935	4	0	0	0	4
1936	3	0	0	0	3
1937	2	0	0	0	2
1938	1	0	0	0	1
1939	0	0	0	0	0

Note: We assume 5 years of studying in university.

Table A2—Cohort-Specific Effects of Jewish Expulsions on Years of Schooling

	(1)	(2)
Percentage of Jews in 1933 × born 1910–1914	−0.3826***	−0.3847***
	(0.1017)	(0.1029)
Percentage of Jews in 1933 × born 1915–1917	−0.4404***	−0.4311***
	(0.0753)	(0.0774)
Percentage of Jews in 1933 × born 1918–1923		−0.1922*
		(0.1144)
R^2	0.216	0.204
Observations	2,385	3,509

Notes: Standard errors clustered by regions are shown in parentheses. The control group is born between 1951 and 1960. Each column controls for region and year of birth fixed effects, gender, and rural dummies.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

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TABLE A3—EFFECTS OF JEWISH EXPULSION ON YEARS OF SCHOOLING BY DIFFERENT COHORTS

	Born between 1910–1923 (1)	Born between 1910–1927 (2)	Born between 1910–1933 (3)	Born between 1910–1938 (4)
Percentage of Jews in 1933 × cohort dummy	−0.3220**	−0.3193**	−0.3008**	−0.2656**
	(0.1583)	(0.1403)	(0.1270)	(0.1249)
Rubble per capita × cohort dummy	0.0025	−0.0048	−0.0001	−0.0037
	(0.0194)	(0.0185)	(0.0166)	(0.0160)
Percentage of refugees × cohort dummy	0.0317	0.0340	0.0384	0.0276
	(0.0308)	(0.0281)	(0.0241)	(0.0238)
Unemployment rate × cohort dummy	0.0387	0.0716	0.0220	0.0319
	(0.0838)	(0.0827)	(0.0740)	(0.0738)
Percentage of votes to Nazi party × cohort dummy	0.0065	0.0008	0.0169	0.0260
	(0.0247)	(0.0197)	(0.0191)	(0.0191)
Percentage of votes to Communist party × cohort dummy	−0.0063	−0.0078	0.0215	0.0320
	(0.0281)	(0.0205)	(0.0216)	(0.0197)
Population size × cohort dummy	−0.0003	−0.0004	−0.0003	−0.0002
	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Income per capita × cohort dummy	−0.0005	−0.0007	−0.0013	−0.0012
	(0.0008)	(0.0009)	(0.0010)	(0.0009)
Urban share × cohort dummy	−0.0020	−0.0021	−0.0068	−0.0133
	(0.0096)	(0.0102)	(0.0116)	(0.0100)
R^2	0.226	0.226	0.225	0.211
Observations	1,951	2,306	2,857	3,406

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1951 and 1960. Each column controls for region and birth year fixed effects. Other controls in each regression are gender and rural dummies and state-cohort trends. Data on unemployment rate, percentage of votes to the Nazi Party and the Communist Party, and population size are from 1932. We also control for the percentage of refugees in 1961 and income per capita in 1929.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE A4—EFFECTS OF JEWISH EXPULSIONS ON YEARS OF SCHOOLING

	(1)	(2)	(3)	(4)
Percentage of Jews in 1933 × born 1910–1927	−0.3129**	−0.2916**	−0.2853**	−0.2942**
	(0.1238)	(0.1174)	(0.1102)	(0.1136)
Percentage of Jews in 1933 × born 1910–1927 × mother has more than basic education		0.0635		−0.3918
		(0.3244)		(0.5126)
Percentage of Jews in 1933 × born 1910–1927 × father has more than basic education			0.4390	0.6461
			(0.3542)	(0.4694)
R^2	0.226	0.309	0.358	0.365
Observations	1,480	1,221	1,214	1,197

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1900 and 1909. Each column is from a separate regression where the main treatment effect varies by parental education in columns 2–4. Each column controls for region and birth year fixed effects, gender, and rural dummies. Columns 2–4 also control for the main effects of parental human capital.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

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TABLE A5—EFFECTS OF DISMISSED TEACHERS ON YEARS OF SCHOOLING–IV RESULTS

	(1)	(2)	(3)	(4)
Dismissed teachers (percent) × born 1910–1927	−0.1450***	−0.1149***	−0.0980***	−0.0942***
	(0.0449)	(0.0394)	(0.0309)	(0.0306)
Dismissed teachers (percent) × born 1910–1927 × mother has more than basic education		0.0588		−0.0376
		(0.0386)		(0.0922)
Dismissed teachers (percent) × born 1910–1927 × father has more than basic education			0.1160	0.1277
			(0.0731)	(0.1020)
R^2	0.155	0.268	0.315	0.332
Observations	1,713	1,480	1,464	1,440
First-stage				
Dependent variable: percentage of dismissed teachers × born 1910–1927				
Percent of Jews in 1933 × born 1910–1927	3.4709***	3.4904***	3.4357***	3.4377***
	(0.2493)	(0.2702)	(0.2743)	(0.2741)
F-statistics	53.51	49.14	42.48	40.76
R^2	0.644	0.664	0.649	0.651
Observations	1,713	1,480	1,464	1,440

Notes: Standard errors clustered by regions are shown in parentheses. The control group is individuals born between 1951 and 1960. Each column is from a separate regression where the main treatment effect varies by parental education in columns 2–4. Each column controls for region and year of birth fixed effects. Columns 2–4 also control for the main effects of parental human capital. Other controls in each regression are gender and rural dummies.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

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